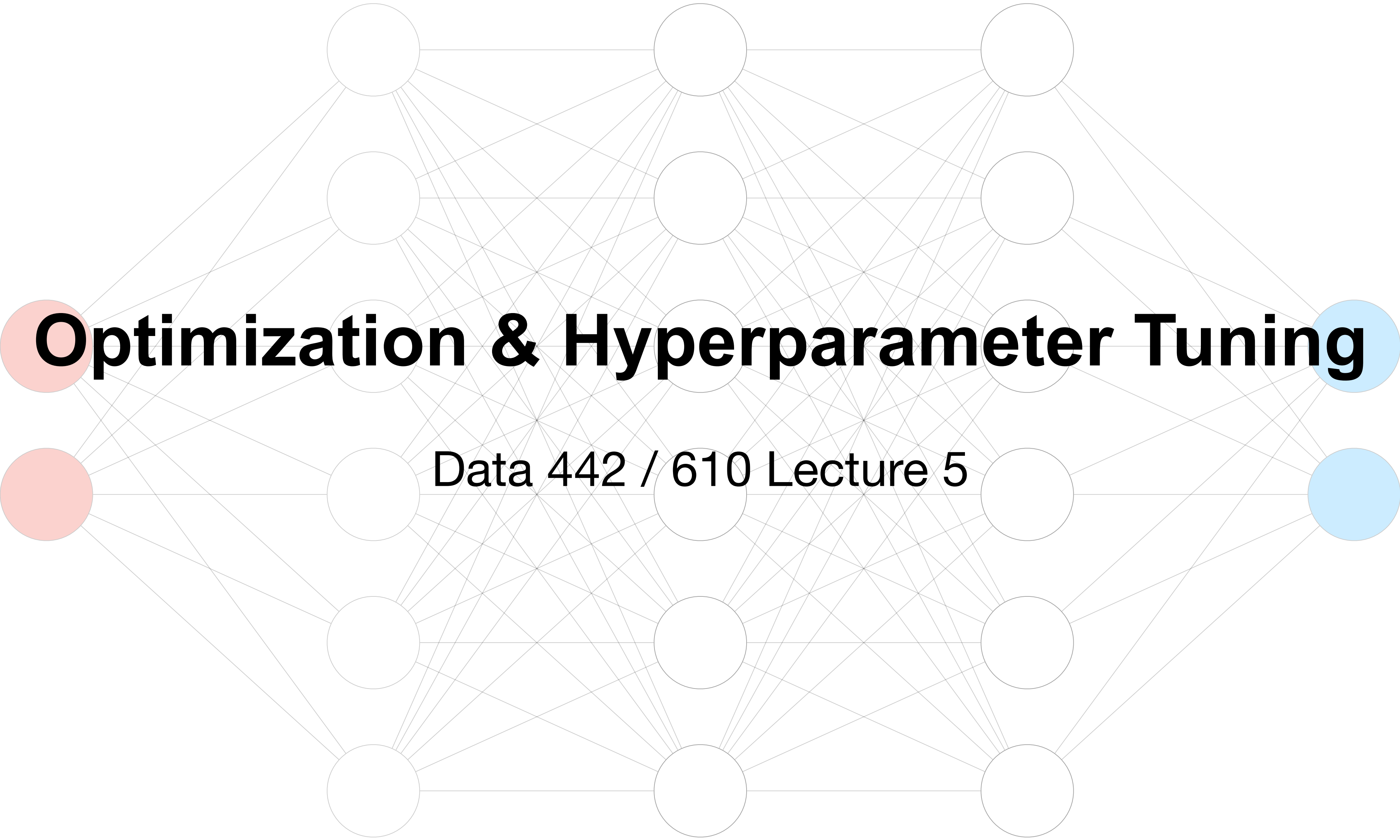




Optimization & Hyperparameter Tuning

Data 442 / 610 Lecture 5

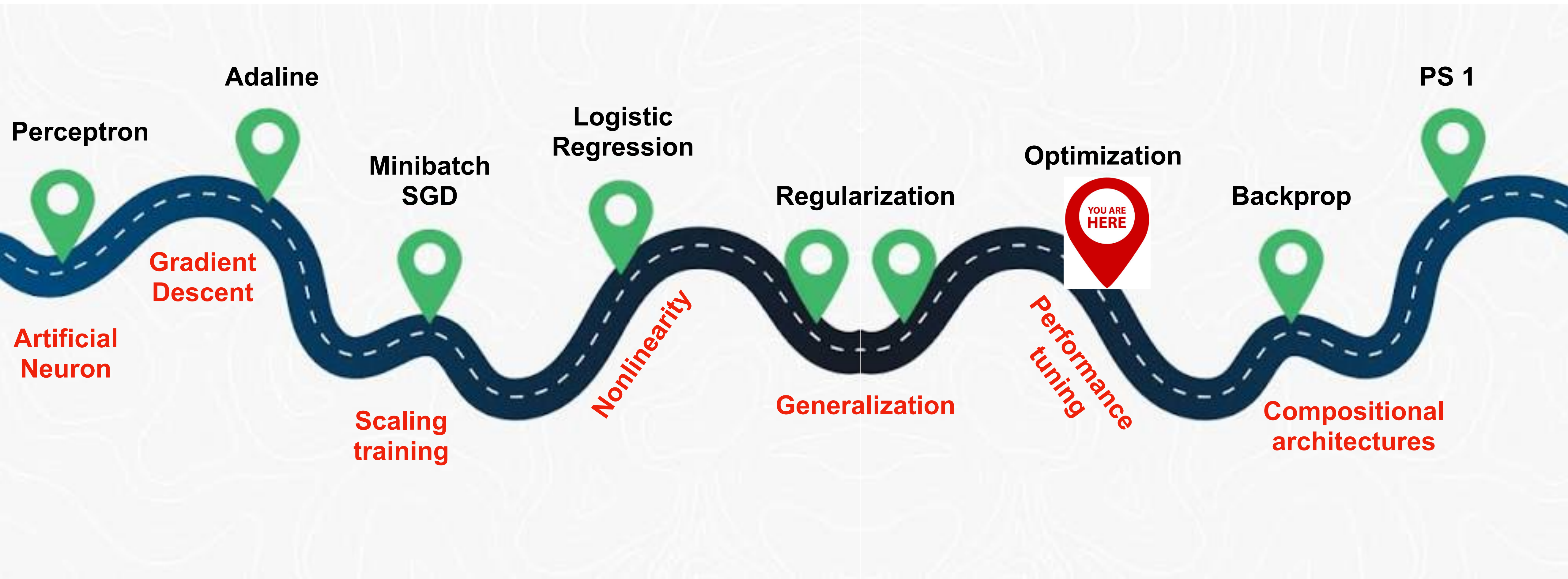
Outline

- Optimization, capacity, and the bias-variance tradeoff
- Hyperparameter Tuning
- Nonlinear optimization

Relevant Reading: *Deep Learning*, Chapter 5, sections 2-4

Bishop, *Pattern Recognition and Deep Learning*, 6.1-6.2 ([link](#))

Roadmap to Deep Learning



Machine learning is complicated

What we have: $X_{\text{train}}, y_{\text{train}}, \mathcal{L} = \mathbb{E} \left[\left(f_{\theta} (X_{\text{train}}) - y_{\text{train}} \right)^2 \right]$

What we want: $\theta^* = \arg \min \mathbb{E} \left[\left(f_{\theta} (X_{\text{test}}) - y_{\text{test}} \right)^2 \right]$

A good ML model will

- Perform well on the training data (no **underfitting**)
- Perform the same on the testing data (no **overfitting**)

ML is an ongoing battle between under- and over-fitting!

Bias and Variance

- A machine learning algorithm is an **estimator** of parameters θ

$$\text{Model: } \hat{y} = f_{\theta}(x) \qquad \text{GD : } (X, y) \rightarrow \hat{\theta}$$

- **Bias** tells us how accurate the estimator is on average

$$\text{Bias}(\hat{\theta}) = \mathbb{E}[\hat{\theta}] - \theta$$

- **Variance** tells us how changes in data affect the estimator

$$\text{Var}(\hat{\theta}) = \mathbb{E}\left[(\hat{\theta} - \mathbb{E}[\hat{\theta}])^2\right]$$

Model capacity

- **Capacity** measures the range of functions a model can fit

$$y = ax^2 + bx + c$$

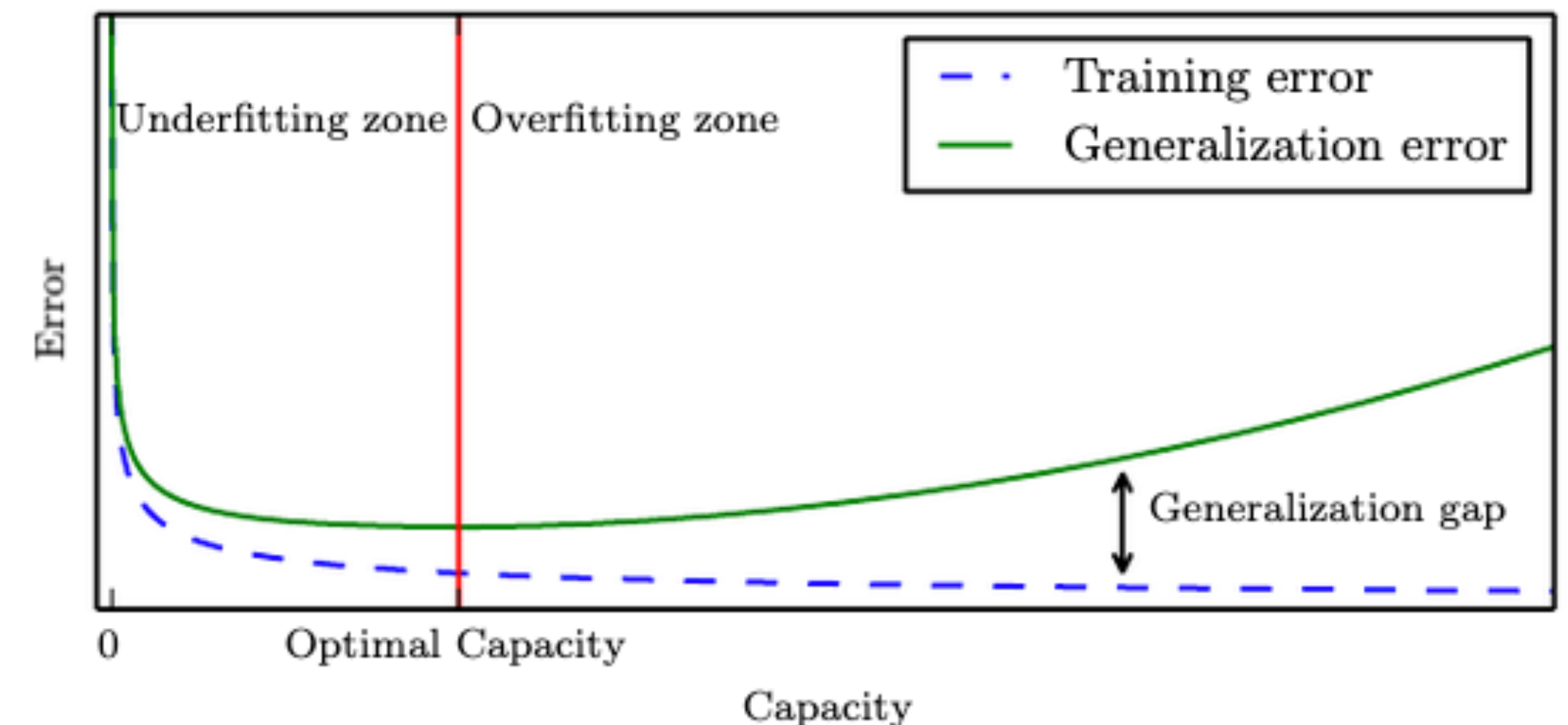
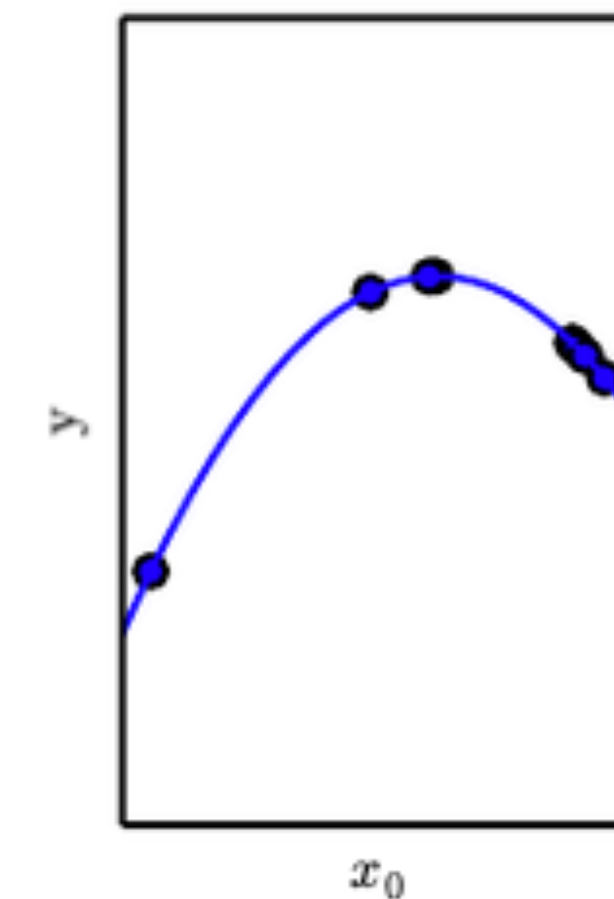
- Models with too much capacity may **overfit**

$$y = \sum_{i=0}^{100} a_i x^i$$

- Models with too little capacity may **underfit**

$$y = mx + b$$

Appropriate capacity



The No Free Lunch Theorem

For any algorithm, any elevated performance over one class of problems is offset by performance over another class

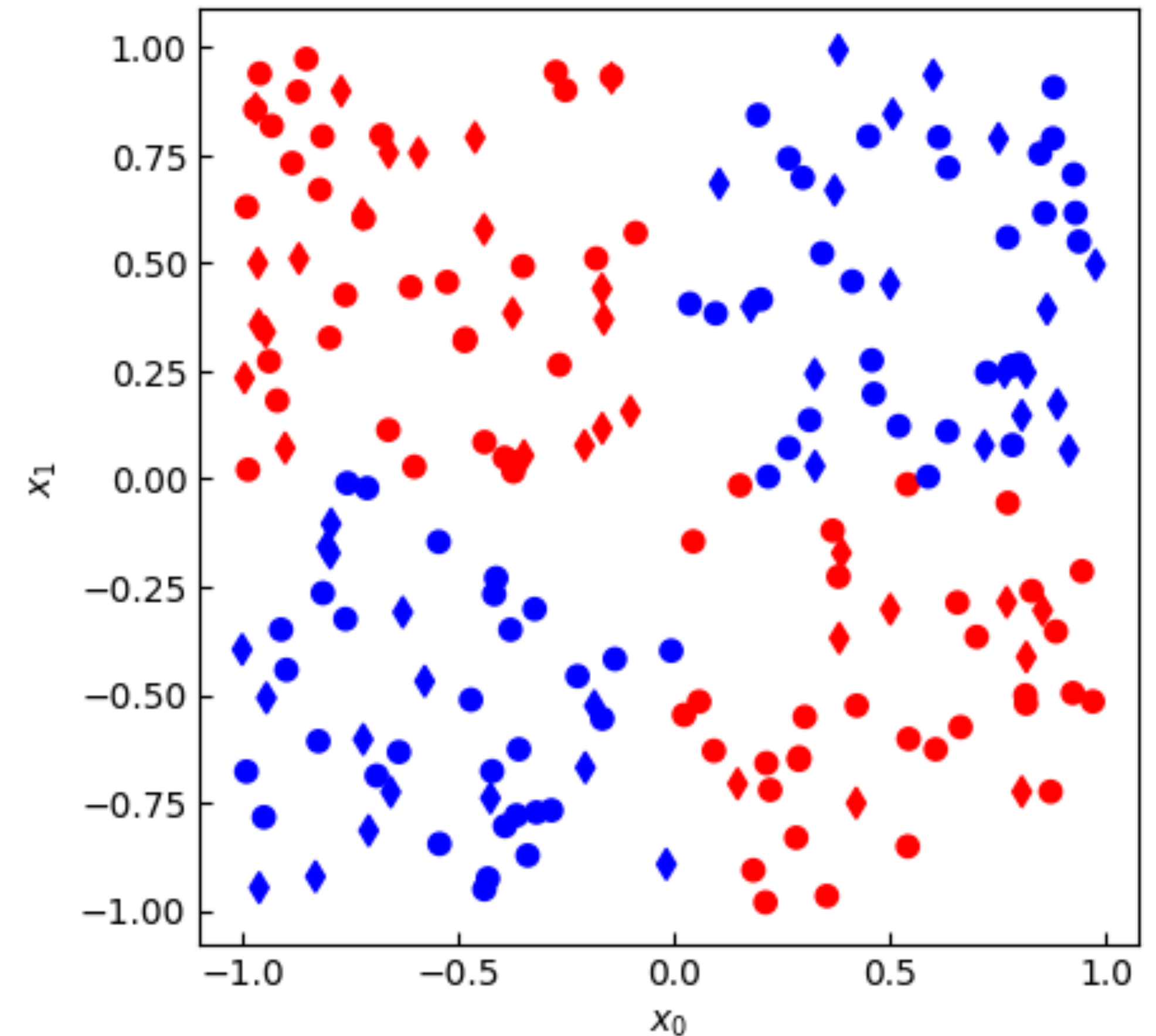
Wolpert & Macready, 1997

- There is no universal best ML algorithm for all problems
- We have to tune our ML approach to each specific problem of interest
- We may benefit from *a priori* knowledge about the structure of possible solutions (**representational capacity**) or the behavior of preferred solutions (**regularization**)

Example: XOR Dataset

Representational capacity: Include polynomial terms like x_0^2 , x_1^2 , x_0x_1 to allow hyperbolic decision surfaces

Regularization: Do not want every polynomial term to contribute, L1 penalty promotes sparsity



Hyperparameter optimization

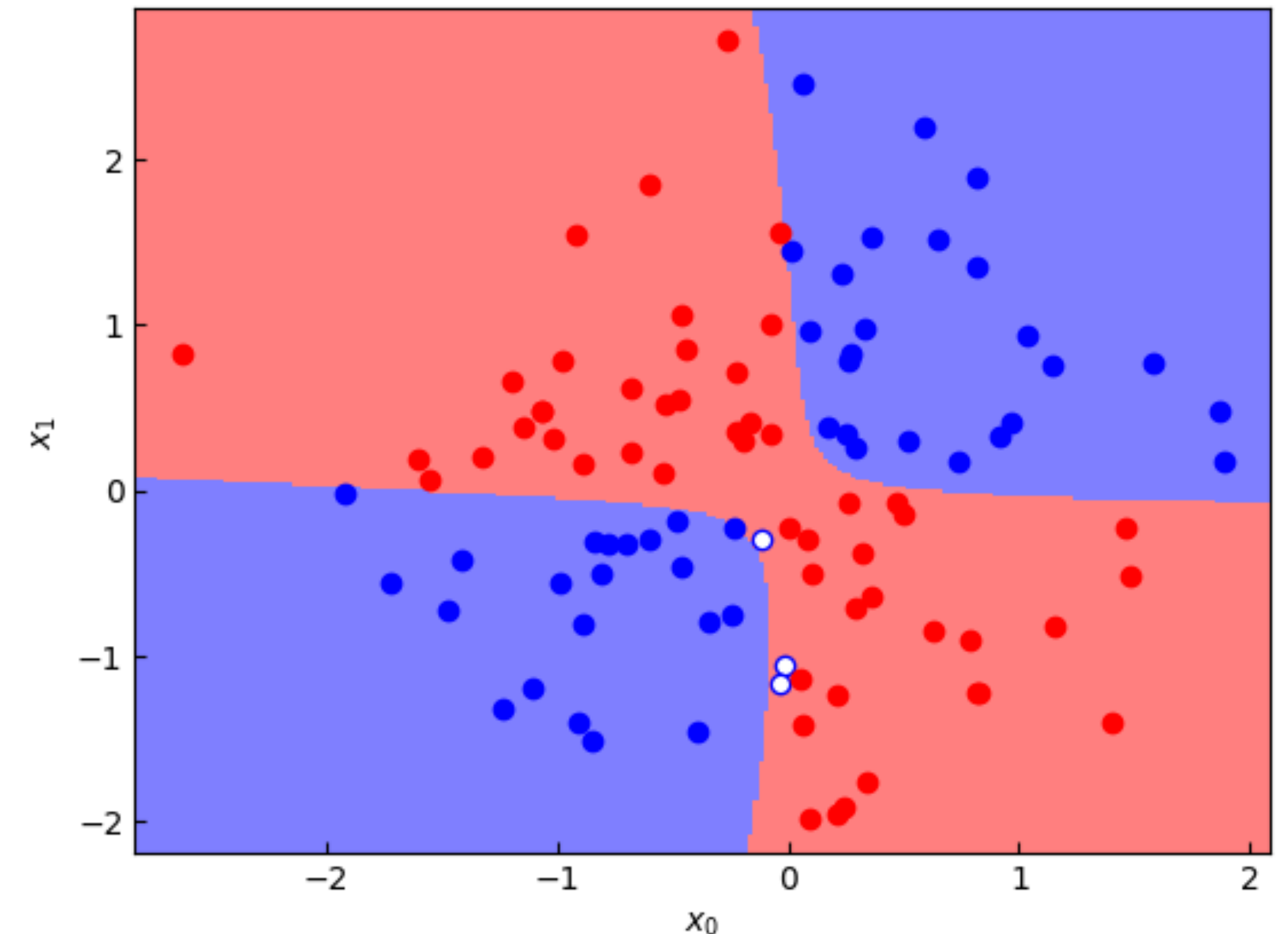
- A **hyperparameter** controls a ML algorithm's behavior but is not directly optimized during training
- Examples include: learning rate, regularization strength, polynomial feature degree
- We will see additional hyper parameters later when we cover more complex models (soon) and learning algorithms (less soon)
- We use **validation sets** and **cross validation** to evaluate the effects of hyperparameters on training and estimate generalization

Hands-on demo: Hyperparameter optimization



Nonlinear optimization

- Optimizing nonlinear functions is challenging because optimization methods for linear functions may no longer apply
- One way around this is to do *linear* optimization on a nonlinear transformation of the inputs $z = \phi(x)$ (polynomial LR)
- Another way is to exploit the **kernel trick**



$$f(x) = b + \sum_i \alpha_i \phi(x)^\top \phi(x^i)$$



$$f(x) = b + \sum_i \alpha_i k_\phi(x, x^i)$$

You Can Just Put Up A Poster At ICML and Nobody Will Stop You

Jimmy Neuron, Chet G. P. Tee IV, Ada Grahd

FedEx is All
You Need

Background

A perennial problem in machine learning research has been how to most efficiently have a poster at ICML. In this work, we show that one can bypass OpenReview via a simple "FedEx trick," similar to yet entirely different from the kernel trick in machine learning. This represents a substantial improvement over the prior SOTA [1].

Research Question: Can you just put up a poster at ICML?
Answer: yes.

Main Contribution

We introduce a novel *FedEx attack* on the conference review system:

- 1) Hastily design a poster in a hotel room.
- 2) Print it and bring to Thursday poster session.
- 3) Surreptitiously mount it on an unused board.

This attack opens new avenues of research into defenses, including embodied human agents fine-tuned to detect and remove spurious content.

Future work

For simplicity, this work treats (and consists of) a poster with only one layer. Future work will focus on deep posters with many layers.

[1] You Can Just Put Up A Poster At AISTATS. JN, CT, AG.
AISTATS 2023.

Theory

Theorem 1 (Free-Lunch Theorem for Posters)

The *Free-Lunch theorem* demonstrates that one can secure a poster at ICML without any discernible work or innovative contribution. Formally, let p denote this poster, and let $\mathcal{P}$ denote the set of posters presented at ICML. This theorem then states $p \in \mathcal{P}$.

The following is a short sketch of the proof. Let $\mathcal{C}$ denote a given review committee, and let $\mathcal{P}$ and $\mathcal{R}$ denote other symbols. Now then, for a given p , we have:

$$\begin{aligned} p &\sim \mathcal{P} \cap (*\mathcal{C}) \\ \Rightarrow \forall p^*, (p, \mathcal{P} - \mathcal{R}[p, \mathcal{P} - \mathcal{R}] &\mapsto p^* \\ \Rightarrow \sum_{i=0}^{|*|} [p_i, (*\mathcal{C})] + \frac{(p_i, \mathcal{P} - \mathcal{R}[p_i, \mathcal{P} - \mathcal{R}]}{|*|} &< |*| \quad (\text{FedEx trick}) \end{aligned}$$

By "x"-ing the previous equation with $(*\mathcal{C})$ once again, we have:

$$\begin{aligned} \frac{(p^*, \mathcal{P} - \mathcal{R}[p^*, \mathcal{P} - \mathcal{R}]}{|*|} &\sim \mathcal{C} \\ \Rightarrow p &\in \mathcal{P} \end{aligned}$$

Your reading this demonstrates this result exactly matches experiment.

As is customary, we also include an additional theorem unrelated to our main contribution:

Theorem 2 (Universal Approximation Theorem for Posters)
Let $\mathcal{P}_w$ denote all subsets of $\{0, 1\}^N$ expressible using posters of width w . Then the following holds for every $U \in \{0, 1\}^N$:

$$\lim_{w \rightarrow \infty} \inf_{P \in \mathcal{P}_w} d_H(P, U) = 0$$

Where d_H denotes the Hamming distance.

Empirics

All experiments were run on a Game Boy Advance™ running Arch Linux, which allows our methods to easily extend to edge computing use cases. We determined the poster width using a hyperparameter grid search with a grid size of 1 (as is standard). Results are shown in Figure 2 below.

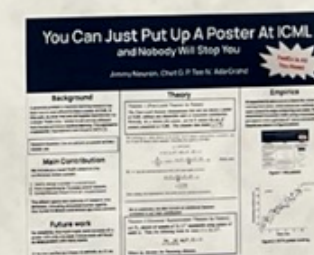


Figure 1: this poster.

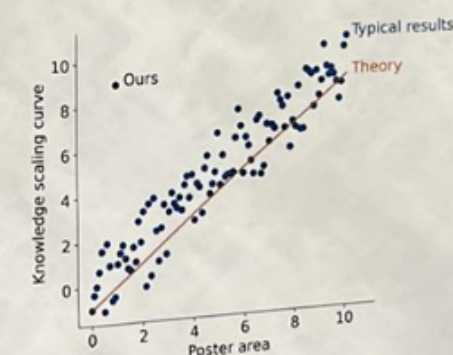


Figure 2: SOTA poster scaling.

A perennial problem in machine learning research has been how to most efficiently have a poster at ICML. In this work, we show that one can bypass OpenReview via a simple "FedEx trick," similar to yet entirely different from the kernel trick in machine learning. This represents a substantial improvement over the prior SOTA [1].

if interested in
collaborating, email
fedex.attack@gmail.com

Hands-on demo: Kernel Methods



That's all!

- Slides will be posted on the course website: <https://colen-teaching.github.io/DATA442-Fall-2026>
- Lab demos will also be posted
- **PS1** due October 2
- **No class Wednesday**
- **No Office Hours Friday** (blackboard/email for any questions)